

UPS OF BUCKMINSTERFULLERENE AND OTHER LARGE CLUSTERS OF CARBON

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Ultraviolet photoelectron spectra (UPS) are reported for mass-selected negative carbon clusters extracted from a pulsed supersonic beam. In the size range from 48 to 84 atoms, three clusters were found to be closed-shell species with appreciable HOMO–LUMO gaps: C_{50} (0.3–0.6 eV), C_{60} (1.5–2.0 eV), and C_{70} (0.7–1.2 eV). UPS data for all other clusters revealed no appreciable HOMO–LUMO gap, indicating they are either open-shell species, or closed-shell species with small HOMO–LUMO gaps. Buckminsterfullerene (C_{60}) was found to have the lowest electron affinity (2.6–2.8 eV) of any cluster. Agreement between these UPS data and electronic structure calculations strongly support the spheroidal shell model for C_{60} .

1. Introduction

Since the original suggestion that C_{60} may have the structure of a truncated icosahedron [1], a flurry of experimental [2–12], and theoretical [13–29] studies have appeared, exploring the possible validity of this proposal and considering its consequences. Although to many the current experimental evidence for such spheroidal carbon shell structures is quite compelling, this evidence still falls somewhat short of a clear proof. To most, such a proof would be spectral, involving high-resolution IR or UV spectroscopy, or (ideally) NMR spectroscopy such as that which so definitively established the structure of dodecahedrane, $C_{20}H_{20}$, the largest previously known polyhedral form of carbon [30]. However, it will be a while yet before enough C_{60} is collected in an NMR sample tube for such an experiment to be feasible.

One step in the direction of spectral probes of C_{60} has been reported recently from this laboratory [6]: a narrow (50 cm^{-1} fwhm) absorption feature near 3860 Å was observed for several van der Waals complexes of C_{60} in a supersonic beam. This absorption feature has been tentatively assigned as the 0–0 band

of the first allowed ${}^1T_{1u} \rightarrow {}^1A_g$ electronic transition which has been predicted to lie in this spectral region [17,24,29]. Although remarkably narrow for any cluster of this size, interaction with the underlying quasicontinuum of vibronic levels broadened this feature sufficiently that all spectral substructure was destroyed – thus preventing detailed confirmation of the assignment.

Here we report another, independent step in the direction of a detailed spectral test of the hypothesis that C_{60} is a truncated icosahedron: ultraviolet photoelectron spectroscopy (UPS) of the mass-selected negative cluster ion. Although still of rather low resolution, these new UPS data provide striking confirmation of electronic structure predictions for icosahedral C_{60} .

2. Experimental

Carbon clusters were prepared by laser vaporization of a graphite disc in a pulsed supersonic nozzle. This cluster source was similar in concept to that used in earlier work in this laboratory on carbon [2–6], as well as other semiconductor materials such as silicon [31,32], and gallium arsenide [33]. In this case the nozzle consisted of a straight 0.2 cm diameter flow tube passing within 0.03 cm of the graphite disc.

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Helium carrier gas with a small concentration of argon (roughly 2.0%) and naphthalene (0.001%) was pulsed through this tube such that at the time of laser vaporization the gas density above the target disc was roughly 1 atm (net gas flow was 1 standard Torr ℓ per pulse in a 300–400 μ s pulse). After a 1.5 cm long passage through this 0.2 cm diameter flow tube past the point of laser vaporization, the carrier gas expanded through a straight 15° cone which, in turn, smoothly joined into a straight cylindrical tube 1 cm inner diameter, 6 cm in length (the “cooling tube”). Gas exiting this final tube freely expanded into a vacuum chamber pumped by three Varian VHS-400 diffusion pumps, and then skimmed by a 15 cm long, 55° included angle skimmer (Beam Dynamics, RU-2) located 25 cm further downstream.

In order to prepare the negative cluster ion beams necessary for mass-selected UPS experiments, two techniques were used. First, and most simply, the carbon cluster source was operated such that the vaporization laser was fired quite early in the carrier gas pulse – thereby allowing some of the laser-induced plasma to survive during the flow through the nozzle. This technique was first introduced by Bloomfield et al. [34]. It readily provides intense charged carbon cluster beams, but as described both by these researchers and by Hahn et al. [7], the cluster ion distribution generally shows little favoritism toward any cluster – let alone C_{60} .

Production of negative cluster beams where the even-numbered clusters were favored and C_{60}^- predominant required the use of a second laser. The intent of this second laser was to regenerate a swarm of free electrons for attachment to the neutral carbon clusters after they had formed and begun to sort out in the early part of the nozzle flow – much as described in an earlier publication from this group [3]. By firing a 1–3 mJ pulse from the fourth harmonic of a Nd:YAG laser (2660 Å) just as the clusters reached the 1 cm diameter “cooling tube”, free electrons were produced by multiphoton ionization in an environment where efficient attachment to the carbon clusters occurred, followed by extensive thermalizing collisions with the carrier gas during passage through the remainder of the 6 cm long cooling tube. In order to aid the production of the required free electron swarm, a small amount of naphthalene was added to the carrier gas (roughly 10 ppm). Pre-

vious experiments in this group have shown there is a very efficient two-photon ionization process in naphthalene at these wavelengths [35]. To further enhance the cooling achieved after formation of the negative cluster ions, a small amount of argon (roughly 2%) was added to the carrier gas as well.

The negative clusters were extracted from the supersonic beam by a high-voltage pulse, directed down a 2 m flight tube, and individual clusters selected by a simple three-grid mass gate [5,32,36]. The selected clusters were then decelerated to 100–150 eV and allowed to drift through the laser detachment region of a time-of-flight UPS spectrometer. This spectrometer was of a new design which features a pulsed, divergent magnetic field of roughly 500 G at the point of photoelectron detachment. This divergent field smoothly mapped onto the parallel field lines of a 2 G solenoidal magnet which surrounded a 2.34 m time-of-flight drift tube. Photoelectrons in such a divergent field are redirected by the “magnetic mirror effect” and their trajectories parallelized such that excellent energy resolution is attained with a collection efficiency of over 98% of the initial photoelectrons. Details of this UPS spectrometer design and its initial applications have appeared elsewhere [37,38].

Calibration of the spectrometer was done by recording the UPS spectrum of Cu^- at a variety of detachment laser energies. All spectra reported below for carbon clusters were taken using an ArF excimer laser (6.4 eV) at a fluence of 1 mJ cm $^{-2}$ to detach the photoelectrons. In the case of C_{60}^- , the photoelectron spectrum was recorded also with the fourth harmonic of the Nd:YAG laser (4.7 eV) to check the instrument calibration and ensure that the spectral features were independent of detachment laser wavelength. UPS spectrum agreed excellently with the ArF excimer laser UPS results described below.

3. Results and discussion

Fig. 1 shows the time-of-flight mass spectrum of the negative carbon cluster distribution prepared by the two-laser technique discussed above. Note that in the 40–80-atom size range, the even clusters are roughly ten times more abundant than the odd, and clusters 50, 60, and 70 are clearly favored. This is

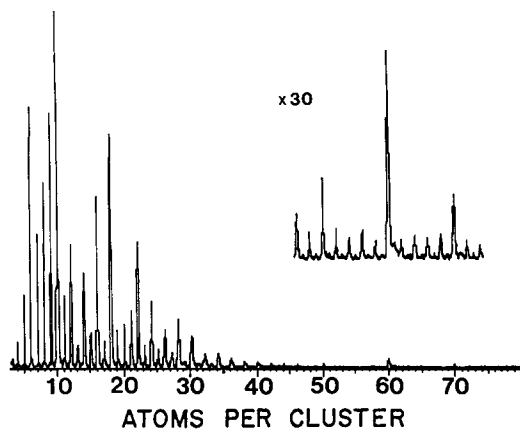


Fig. 1. Time-of-flight mass spectrum of negative carbon cluster ions prepared in a pulsed supersonic beam by a two-laser technique. The first laser was used to generate carbon vapor by irradiating a graphite disc, the second (fourth harmonic of Nd:YAG) was used to generate a shower of electrons for attachment to the neutral clusters as they emerged into a downstream "cooling tube" prior to free expansion.

essentially the same distribution published by Liu et al. [3] in earlier work on another apparatus in this laboratory using much the same technique.

Using the mass gate, individual cluster sizes from this distribution were selected and the UPS spectrum recorded. Fig. 2 presents a broad overview of the results for all even-numbered clusters in the 48–72-atom size range, as well as C_{80}^- and C_{84}^- . The concentrations of the odd-numbered clusters in this cluster distribution were too low to permit study. With the exception of C_{60}^- (and to a lesser extent C_{70}^- and C_{50}^-), all carbon clusters in this size range showed only a broad, sloping photoelectron spectrum. But for 50, 60 and 70 there a separate feature is seen near threshold. This initial bump in the UPS spectrum is by far the most prominent and well separated for the 60th cluster. A larger view of this C_{60}^- UPS spectrum is presented in fig. 3.

As we have discussed previously [37,38], the UPS pattern shown in fig. 3 is characteristic of a molecule having a closed-shell electronic structure for the neutral species with a large HOMO-LUMO gap. For such a molecule the extra electron added to form the negative ion must lie alone in the LUMO, and it is the detachment of this lone electron that occurs at threshold. The next most deeply bound electrons lie in what corresponded to the HOMO of the neutral

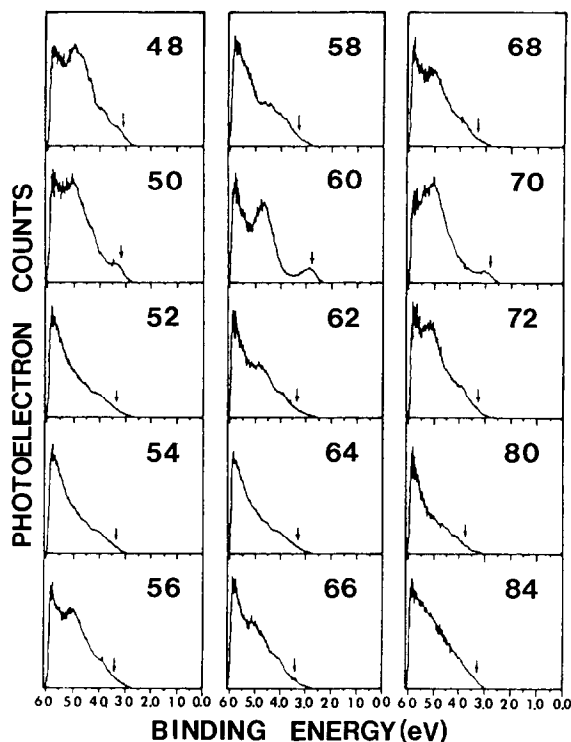


Fig. 2. Ultraviolet photoelectron spectra of negative carbon cluster ions taken with an ArF excimer laser (6.4 eV). The ions were mass selected from the cluster ion beam whose distribution is shown in fig. 1. Arrows indicate the estimated vertical electron affinities of the corresponding neutral clusters, corrected for thermal effects.

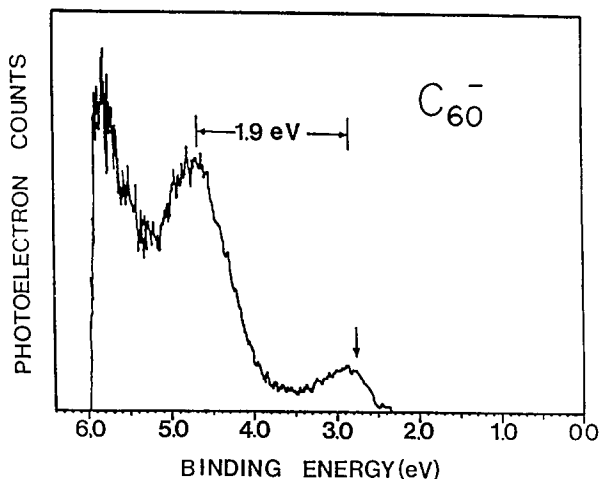


Fig. 3. Expanded view of the ultraviolet photoelectron spectrum of C_{60}^- (Buckminsterfullerene) taken with an ArF excimer laser.

molecule, and, generally for a cluster of this size, there will be many such electrons – either because of orbital degeneracy, or the presence of several orbitals close in energy. The UPS pattern for the negative ion of such a closed-shell molecule will then be characterized by a small initial feature followed by a gap (the HOMO–LUMO gap) followed by a much larger feature, just as in fig. 3 for C_{60}^- .

In the case of fig. 3, the resolution is limited not by the spectrometer, but by the molecule itself. By varying carrier gas composition and sampling clusters at different flight times in the supersonic beam, it was clear that the dominant contribution to the spectral widths seen in fig. 3 is still thermal. Without further cooling it is impossible to be precise in extraction of electron affinities and HOMO–LUMO gaps from these data. However, comparison to similar features for copper cluster UPS [38,40,41], where electron affinities are known by independent measures of the one-photon detachment threshold [36], suggests a reasonable estimate for the electron affinity of C_{60} is 2.6–2.8 eV as shown by the arrow in fig. 3. Similar estimates for the other carbon clusters are indicated by arrows in fig. 2. It is clear from the data that C_{60}^- has the lowest vertical electron affinity and largest apparent HOMO–LUMO gap of any cluster in this region, and C_{70}^- is close behind.

3.1. Other valence isomers

One of the most dramatic results of this study is the strong dependence of the photoelectron spectrum on the cluster formation history. As discussed in section 2, negative carbon cluster beams can be produced using the vaporization laser alone. Fig. 4 shows the cluster distribution obtained by this technique. Note that now there is only a small extra abundance seen for C_{60}^- as compared to the other even clusters, and the odd clusters are now almost as intense as the even. Under these conditions the UPS patterns for various clusters were found to be virtually superimposable. Fig. 5 shows such data for C_{58} , C_{59} , and C_{60} . The only significant difference seen here is in C_{60} , where the pattern is a superposition of that seen for 58 and 59 with a small amount of the pattern seen in fig. 3. In fact, our first UPS experiments on carbon clusters done several months before the ones shown here gave *exactly* the same UPS pattern

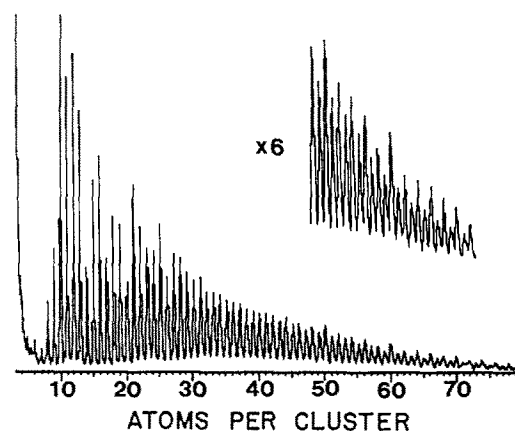


Fig. 4. Time-of-flight mass spectrum of negative carbon clusters prepared in a supersonic beam using only the primary vaporization laser to provide the source of ionization. Note that C_{60} is not particularly special under these conditions.

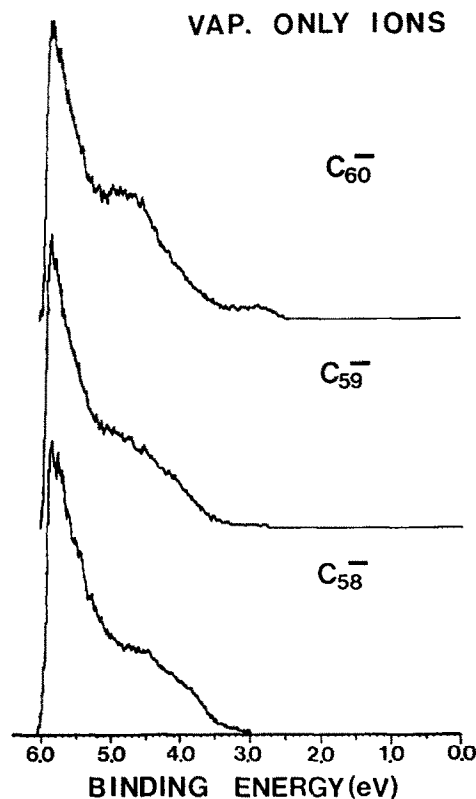


Fig. 5. Ultraviolet photoelectron spectra of C_{58}^- , C_{59}^- , and C_{60}^- taken with an ArF excimer laser. These clusters were extracted from the supersonic beam whose cluster distribution is shown in fig. 4.

for all clusters, including C_{60} . These early experiments were done with a less intense cluster expansion source which produced a negative carbon cluster distribution with no prominence of C_{60} whatsoever.

There appear, therefore, to be at least two distinct forms of these negative carbon clusters. One form (of which there may be many varieties) is made early on in the cluster formation process, has an open-shell electronic structure, and shows no preference for one cluster size over another. The second form, which is made only after extensive reaction and annealing processes in the nozzle, has (at least for 50, 60, and 70) a closed-shell electronic structure, and is far more stable for C_{60} than any other cluster size.

4. Conclusions

These UPS results, including the remarkable behavior as a function of cluster source conditions, are well in accord with the spheroidal carbon shell model as it has been originally proposed and developed over the past few years [2–6]. In the case of C_{60} , the measured 2.6–2.8 eV electron affinity and 1.5–2.0 eV HOMO–LUMO gap is in excellent agreement with most theoretical predictions for the truncated icosahedral structure – particularly the most recent calculations of Rosén and co-workers [29]. Further UPS studies are in progress using higher detachment laser photon energies, and extending down to the smallest clusters where initial survey experiments have already revealed strikingly rich and well-resolved spectral structure which should provide stringent tests for any models of the structure of these fascinating molecules.

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